

Frobenius 1

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Problem 1. Let V be a finite dim vector space over a field F . Construct a natural map $tr: Hom_F(V, V) \rightarrow F$. (the domain means F -linear maps). Let $GL(V)$ act on $Hom_F(V, V)$ by conjugation. Show tr is constant on $GL(V)$ orbits. Now let finite G act on linearly on V (ie V is a finite dim rep of G). Explain how we get a map $tr_V: G \rightarrow F$, and why this factors through $G \rightarrow G/G :=$ the set of conjugacy classes (the quotient of G by G acting on itself by conjugation).

Claim 1a. Construct the map $tr_V: Hom_F(V, V) \rightarrow F$

Proof. Choose a basis for V , thus there exists a isomorphism $\beta: Hom_F(V, V) \rightarrow M_{n \times n}$ where $n = \dim V$. $tr_{M_{n \times n}}$ is the standard trace, ie sum of diagonal entries. Thus $tr_V = tr_{M_{n \times n}} \circ \beta$. $\square$

Claim 1b. Let $GL(V)$ act on $Hom_F(V, V)$ by conjugation. Show tr is constant on $GL(V)$ orbits.

Proof. By Claim 1d, $\forall A, B \in M_{n \times n}$, $tr(AB) = tr(A)tr(B)$. Thus $\forall M \in GL(V)$, $\forall f \in Hom_F(V, V)$, thus $tr(MfM^{-1}) = tr(MM^{-1}f) = tr(I)f = tr(f)$. Thus tr is constant on $GL(V)$ orbits. $\square$

Claim 1c. Explain how we get a map $\chi_V: G \rightarrow F$, and why this factors through $G \rightarrow G/G :=$ the set of conjugacy classes

Proof. $\chi_V: G \rightarrow F$ where $\chi = tr \circ \rho$ where $\rho: G \rightarrow GL(V)$ is the representation. Since $\forall g, h \in G$, $\chi_V(ghg^{-1}) = tr(\rho(g)\rho(h)\rho(g)^{-1}) = tr(\rho(h)) = \chi_V(h)$. Thus by the prehomework for semester 1, χ factors through π the map to G/G where g is mapped to the conjugation class. Because there exists $\phi: G/G \rightarrow F$, where $[g] \rightarrow \chi(g)$. Thus $\chi_V = \phi \circ \pi$. $\square$

Claim 1d. For $A, B \in GL_n(F)$, $tr(AB) = tr(BA)$

Proof. $tr(AB) = \sum_{i=1}^n \sum_{j=1}^n A_{i,j} B_{j,i}$ and $tr(BA) = \sum_{k=1}^n \sum_{l=1}^n B_{k,l} A_{l,k} = \sum_{l=1}^n \sum_{k=1}^n A_{l,k} B_{k,l} = \sum_{i=1}^n \sum_{j=1}^n A_{i,j} B_{j,i} = tr(AB)$. Thus $tr(AB) = tr(BA)$ $\square$

Problem 2. Suppose G acts (linearly) on V and W . Define what we should mean by $Hom_G(V, W)$ – F -linear maps that commute with the G -action. And then what we should mean by an isomorphism between two reps of G . Show that if two reps V, W are isomorphic, then the function tr_V and tr_W are equal. A corollary of the Frobenius theorem will be that for $F = \mathbb{C}$ the converse holds – two finite dim complex reps of finite G are iso iff their trace functions are the same.

Proof. When $f_1 \cong f_2$, $\exists \alpha \in Hom_G(V, W)$, be the isomorphism of G -representations. Thus $\forall g \in G$, $\alpha \circ f_1(g) = f_2(g) \circ \alpha \implies \alpha \circ f_1(g) \circ \alpha^{-1} = f_2(g)$. Thus the matrices, $f_1(g)$ and $f_2(g)$ are conjugate.

Suppose $a: G \rightarrow GL(V)$ and $b: G \rightarrow GL(W)$ are representations of G which are isomorphic. Thus $\exists \alpha \in Hom_G(V, W)$ which is a isomorphism. By 2, we know $\forall g \in G$, $\alpha \circ a(g) = b(g) \circ \alpha \implies$

$\alpha \circ a(g) \circ \alpha^{-1} = b(g)$, in other words isomorphic representations are conjugate, equivalently related by change of basis. $\forall g \in G$, $\chi_b(g) = \text{tr}(b(g)) = \text{tr}(\alpha \circ a(g) \circ \alpha^{-1}) = \text{tr}(\alpha \circ \alpha^{-1} \circ a(g))$, by 3a.ii. Thus $\chi_b(g) = \text{tr}(b(g)) = \text{tr}(\alpha \circ \alpha^{-1} \circ a(g)) = \text{tr}(a(g)) = \chi_a(g)$. Thus $\chi_b = \chi_a$. $\square$

Problem 3. Suppose G acts on V and W . Define an action on $\text{Hom}_F(V, W)$. Explain why $V \oplus W$ is like a sum, and $\text{Hom}_F(V, W)$ is like a product

Proof. $V \oplus W$ is like a sum because the dimensions add up thus $\dim V \oplus W = \dim V + \dim W - \dim V \cap W$. Assuming $\dim V \cap W = 0$, $\dim V + \dim W = \dim V \oplus W$.

There exists $\alpha : \mathbb{F}^{\dim V} \rightarrow V$ and $\beta : \mathbb{F}^{\dim W} \rightarrow W$ be isomorphisms. Thus there exists a isomorphism from $\text{Hom}_F(V, W) \rightarrow M_{\dim W \times \dim V}$ where $f \rightarrow \beta^{-1} \circ f \circ \alpha$.

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ \uparrow \alpha & & \uparrow \beta \\ \mathbb{F}^{\dim V} & \xrightarrow{\beta^{-1} \circ f \circ \alpha} & \mathbb{F}^{\dim W} \end{array} .$$

$$\text{Hom}_{\mathbb{F}}(V, W) \cong M_{\dim W \times \dim V} . \dim \text{Hom}_{\mathbb{F}}(V, W) = \dim M_{\dim W \times \dim V} = \dim V \dim W . \quad \square$$

Problem 4. Prove that $\text{tr}_{V \oplus W} = \text{tr}_V + \text{tr}_W$, and $\text{tr}_{\text{Hom}(V, W)} = \text{tr}_V \bar{\text{tr}}_W$ (the product of tr_V and the complex conjugate of tr_W .) For the second one explain why it is enough to prove the following: Let A be $n \times n$ and B $m \times m$, consider $L_A \circ R_B : M_{n \times m} \rightarrow M_{n \times m}$ sending $M \rightarrow AMB$. Show that if X is an eigen column for A , value a , and Y^t an eigen row for B , value b , then XY^t is an eigenmatrix for the operator, with value ab . Show that if $\{X_i\}$ is a basis of columns and $\{Y_j^t\}$ a basis of rows then $\{X_i Y_j^t\}$ is a basis for $M_{n \times m}$. Conclude that if A and B are diagonalizable so is this operator, and its eigenvalues are $\{a_i b_j\}$. Show in this case its trace is $\text{tr}_A \text{tr}_B$. Note $n \times n$ matrices is naturally $\mathbb{R}^{2n^2}$. It has an obvious topology, we can do calculus on it. Explain why diagonalizable $n \times n$ matrices are dense in all matrices – you may want to use the fact that every square matrix over the complex numbers is conjugate to an upper triangular matrix. Now explain why if A_i are matrices with limit A , and B_i matrices with limit B , then the operators for A_i, B_i converge to the operators for A, B – explain what converge means. Now prove that for any A and B , the operator above has trace $\text{Tr}(A)\text{Tr}(B)$

Proof. $P := V \oplus W$ where V and W are separate vector spaces which G acts on and thus are G -invariant in P . G acts on P where $g \rightarrow \begin{bmatrix} g_V & 0 \\ 0 & g_W \end{bmatrix}$ where $g_V \in GL(V)$ and $g_W \in GL(W)$ are the representations of V and W . Thus $\forall g \in G$ $\text{tr}_{V \oplus W}(g) = \text{tr}_V(g_V) + \text{tr}_W(g_W)$, thus $\text{tr}_{V \oplus W} = \text{tr}_V + \text{tr}_W$. $\square$

Claim 4b. Show that if X is an eigen column for A , value a , and Y^t an eigen row for B , value b , then XY^t is an eigenmatrix for the operator, with value ab .

Proof. $AXY^t B = aXY^t b = abXY^t$. Thus XY^t is the eigenmatrix of value of ab . $\square$

Claim 4c. Show that if $\{X_i\}$ is a basis of columns for F^m and $\{Y_j^t\}$ a for $(F^n)^T$ basis of rows then $\{X_i Y_j^t\}$ is a basis for $M_{n \times m}$

Proof. $|\{X_i Y_j^T\}| = |\{X_i\}| |\{Y_j^T\}| = m \cdot n = \dim(M_{n \times m})$. $0 = \sum_{i=1}^n \sum_{j=1}^m \alpha_{i,j} X_i Y_j^T = \sum_{i=1}^n X_i \sum_{j=1}^m \alpha_{i,j} Y_j^T$. Let $b_i = \sum_{j=1}^m \alpha_{i,j} Y_j^T$, where b_i is a row vector. $\sum_{i=1}^n X_i b_i = 0$ is the 0 matrix and thus the only way for $(\sum_{i=1}^n X_i b_i)_q$ or the q th column of the matrix is 0. $(\sum_{i=1}^n X_i b_i)_q = (\sum_{i=1}^n X_i b_i) e_q = 0$ where e_q is the q th standard basis for F^m . $0 = (\sum_{i=1}^n X_i b_i) e_q = \sum_{i=1}^n X_i b_i e_q = 0$. Since b_i is a row vector and e_q is a column vector thus $b_i e_q \in F$ for all q . Thus $\sum_{i=1}^n X_i b_i e_q = \sum_{i=1}^n b_i e_q X_i = 0 \implies b_i e_q = 0$ since X_i is a basis. $b_i e_q = 0$ for all q thus every component of b_i is 0. Thus $\sum_{j=1}^m \alpha_{i,j} Y_j^T = 0 \implies \alpha_{i,j} = 0$ because Y_j is a basis and thus linearly independent. Thus $X_i Y_j^T$ is a basis for $M_{n \times m}$ $\square$

Claim 4d. Conclude that if A and B are diagonalizable so is this operator, and its eigenvalues are $\{a_i b_j\}$.

Proof. Since A is diagonalizable, exists a eigenbasis $\{X_i\}$ with eigenvalues a_i . Since B is diagonalizable, there exists a eigenbasis $\{Y_j^T\}$ with eigenvalues b_j . Thus by problem 4, $A X_i Y_j^T B = a_i b_j X_i Y_j^T$. Thus the eigenvalues of the operator is $\{a_i b_j\}$. $\square$

Claim 4d. Show in this case its trace is $\text{tr}_A \text{tr}_B$.

Proof. $\text{tr}_A = \sum_{i=1}^n a_i$ and $\text{tr}_B = \sum_{j=1}^m b_j$. Thus $\text{tr}_{L_A \circ R_B} = \sum_{i=1}^n \sum_{j=1}^m a_i b_j = \sum_{i=1}^n a_i \sum_{j=1}^m b_j = (\sum_{i=1}^n a_i) (\sum_{j=1}^m b_j) = \text{tr}_A \text{tr}_B$. $\square$

Claim 4e. Explain why diagonalizable $n \times n$ matrices are dense in all matrices – you may want to use the fact that every square matrix over the complex numbers is conjugate to an upper triangular matrix.

Proof. By the fact, $\forall A \in M_{n \times n}$ is conjugate to an upper triangular matrix. Choose a basis where A is

upper triangular thus $A = \begin{bmatrix} \lambda_1 & * & * & \dots \\ & \lambda_2 & * & \dots \\ & & \lambda_3 & \dots \\ & & & \ddots \end{bmatrix}$. Create a matrix D where all the off diagonal entries

are A_{ij} where $i \neq j$. If $\lambda_i = \lambda_j$ in the $D_{i,i} = \lambda_i - \alpha_i$ and $D_{j,j} = \lambda_j - \alpha_j$. Choose α_i such that

$\forall i, j \in \mathbb{N}, D_{i,i} \neq D_{j,j}$. Thus $D = \begin{bmatrix} \lambda_1 - \alpha_1 & * & * & \dots \\ & \lambda_2 - \alpha_2 & * & \dots \\ & & \lambda_3 - \alpha_3 & \dots \\ & & & \ddots \end{bmatrix}$. The characteristic polynomial,

$\chi_D(x) = \prod_{i=1}^n (\lambda_i - \alpha_i - x)$ since α_i was chosen such that $\lambda_i - \alpha_i \neq \lambda_j - \alpha_j$ where $i \neq j$. χ_D has unique

roots thus D is diagonalizable. $|A - D|^2 = \left\| \begin{bmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \alpha_3 & \\ & & & \ddots \end{bmatrix} \right\|^2 < \epsilon \implies \alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \dots < \epsilon$.

For $\epsilon < 1$. Choose $\alpha_i = (\frac{\epsilon}{2})^i$. Thus $\sum_i \alpha_i^2 = \sum_{i=1}^n ((\frac{\epsilon}{2})^2)^i = \frac{1 - (\frac{\epsilon}{2})^{2n}}{1 - (\frac{\epsilon}{2})^2} - (\frac{\epsilon}{2})^2 < \epsilon$. For $\epsilon > 1$. Choose $\alpha_i = (\frac{1}{2})^i$. Thus $\sum_i \alpha_i^2 < \epsilon$. Thus $|A - D| < \epsilon$. where D is diagonalizable thus diagonalizable matrices are dense in all matrices. $\square$

Claim 4f. Now explain why if A_i are matrices with limit A , and B_i matrices with limit B , then the operators for A_i, B_i converge to the operators for A, B – explain what converge means.

Proof. $\|(L_{A_i} \circ R_{B_i})(M) - (L_A \circ R_B)(M)\| = \|A_i M B_i - A M B\| = \|(A_i - A)M(B_i - B)\| = \|A_i - A\| \|M\| \|B_i - B\|$ because $\forall C, D \in M_{n \times n}, \|CD\| \leq \|C\| \|D\|$. Thus $\|(L_{A_i} \circ R_{B_i})(M) - (L_A \circ R_B)(M)\| = \|A_i - A\| \|M\| \|B_i - B\| \leq \epsilon^2 \|M\|$. Let $\epsilon_o = \epsilon^2 \|M\|$. Thus $\|(L_{A_i} \circ R_{B_i})(M) - (L_A \circ R_B)(M)\| \leq \epsilon_o$. As $i \rightarrow \infty$ then $\epsilon_o \rightarrow 0$. Thus $L_{A_i} \circ R_{B_i}$ converges to $L_A \circ R_B$. Converge means for some choice of $\epsilon_o \exists i \in \mathbb{N} \ni \forall m \in \mathbb{N} \ni m \geq i, \|(L_{A_m} \circ R_{B_m})(M) - (L_A \circ R_B)(M)\| < \epsilon_o$. $\square$

Claim 4g. Now prove that for any A and B , the operator above has trace $\text{Tr}(A)\text{Tr}(B)$

Proof. $\text{tr}(L_A \circ R_B) = \lim_{m \rightarrow \infty} (\text{tr}(L_{A_m} \circ R_{B_m})) = \lim_{m \rightarrow \infty} (\text{tr}(A_i)\text{tr}(B_i))$ (by the previous claim). Thus $\text{tr}(L_A \circ R_B) = \lim_{m \rightarrow \infty} (\text{tr}(A_i)\text{tr}(B_i)) = \text{tr}(A)\text{tr}(B)$. $\square$

Problem 5. Let Q be a finite set. Explain why $\text{Hom}_{\text{sets}}(Q, F)$ is a vector space of $\dim |Q|$, and why its reasonable to call it Q^F . Give a natural basis. Say F is a subfield of C . Define a natural positive def F -valued hermitian inner product, so that the constant function 1 has norm 1. In the next homework we will prove the main step in Frobenius's theorem, showing that characters of irreducible representations are orthonormal w/r to this natural inner product.

Proof. $\forall \alpha, \beta \in \text{Hom}_{\text{sets}}(Q, F)$ and. Let $Q = \{q_i | i \in \mathbb{N}_n\}$. $\alpha := \text{Hom}_{\text{sets}}(Q, F) \rightarrow F^{|Q|}$ where $f \rightarrow \begin{bmatrix} f(q_1) \\ \vdots \\ f(q_n) \end{bmatrix}$. Let $\beta := F^{|Q|} \rightarrow \text{Hom}_{\text{sets}}(Q, F)$ where $(f_1, \dots, f_n) \rightarrow [q_i \rightarrow f_i]$. $\forall (f_1, \dots, f_n) \in F^{|Q|}, (\alpha \circ \beta)(f_1, \dots, f_n) = \alpha([q_i \rightarrow f_i]) = (f_1, \dots, f_n)$. $\forall f \in \text{Hom}_{\text{sets}}(Q, F), (\beta \circ \alpha)(f) = \beta((f(q_1), \dots, f(q_n))) = [q_i \rightarrow f(q_i)] = f$. Thus α is a bijection. $\forall f, g \in \text{Hom}_{\text{sets}}(Q, F), \forall a, b \in F, \alpha(af + bg) = ((af + bg)(q_1), \dots, (af + bg)(q_n)) = (a(f(q_1), \dots, f(q_n)) + b(g(q_1), \dots, g(q_n)))$. Thus it is vector of $|Q|$ elements thus $\dim \text{Hom}_{\text{sets}}(Q, F) = |Q|$. It is reasonable to call it F^Q as it has elements of F in Q slots. Let $Q_i \in \text{Hom}_{\text{sets}}(Q, F)$ be the function which sends i th element of Q to 1 in F and everything else to 0.

Thus define $\angle$, a hermitian dot product where $\forall f, g \in \text{Hom}_{\text{sets}}(Q, F), \langle f, g \rangle = \frac{1}{|Q|} \sum_{i=1}^{|Q|} f(q_i)g(\bar{q}_i)$. Thus $\langle 1, 1 \rangle = \frac{1}{|Q|} \sum_{i=1}^{|Q|} 1 = 1$. $\square$